

Anqi Dai

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Data Science undergraduate student with a strong background in statistics, machine learning, and systems programming. I am especially interested in applying AI and data science to interactive systems such as games and intelligent interfaces.

Education

University College London (UCL) 2026 – 2027

MSc in Data Science and Machine Learning

University of Michigan, Ann Arbor 2023 – 2026

Bachelor of Science in Data Science

- Relevant Coursework: Data Structures and Algorithms; Statistics and Artificial Intelligence; Linear Algebra; Regression Analysis; Machine Learning; Web Systems; Computational Methods.
- GPA: 3.75/4.0
- Honors: University Honors (Fall 2023, Winter 2024, Fall 2024, Winter 2025)

Data Science Projects

Social Media-Based Mental Health NLP Retrieval System 2026.01 – Present

Designed a lightweight, non-clinical NLP retrieval framework to provide empathetic mental health support by mining social media texts.

- Developed keyword retrieval and context analysis algorithms to extract semantic features from user comments and posts, reducing ambiguity in natural language processing.
- Built multi-dimensional evaluation models to predict users' potential mental distress (e.g., anxiety, depression) and MBTI personality traits without clinical diagnosis, generating highly personalized recommendations.

Image Classification Project (Python, PyTorch) 2025.11

Trained image-classification models to distinguish ten dog breeds on a dataset of 8,000+ labeled images, comparing convolutional neural networks with a lightweight Vision Transformer.

- Implemented CNNs with batch normalization, dropout, early stopping, and L2 regularization to study training dynamics, overfitting, and parameter efficiency.
- Built a data augmentation pipeline (flips, rotations, color jitter, affine transforms) to improve generalization and reduce dataset bias.
- Implemented a compact ViT with patch embedding, positional encoding, multi-head self-attention, and MLP blocks; compared its performance to CNN baselines.
- Achieved strong validation AUROC and performed error analysis revealing class imbalance and feature biases affecting test generalization.

Clinical Mortality Prediction (Python, scikit-learn) 2025.09

Built models to predict in-hospital mortality using ICU data from the first 48 hours, based on time-stamped vitals, lab results, and key demographic and admission information.

- Processed clinical time series into usable features, including summary statistics and one-hot-encoding.
- Trained logistic regression and RBF SVM models with stratified cross-validation, optimizing AUROC under class imbalance.

- Applied cost-sensitive training and tuned decision thresholds to improve F1-score without reducing AUROC.
- Examined model coefficients and compared linear, ridge, and kernel-based approaches to balance performance and interpretability.

Michigan Data Science Team (MDST)

2024.09 – 2025.05

Spotify Music Analytics (Project Lead)

- Analyzed Spotify listening records via API; identified user listening trends and visualized preferences using bar charts, word clouds, and other tools.
- Designed algorithms to recommend personalized playlists; team evaluation score averaged 8.7/10.
- Used SQL for database management and analyzed music preference differences across user groups.

C++ Programming Projects

Algorithm Games

2025.01 – 2025.04

A collection of algorithm-focused mini-games demonstrating understanding of data structures.

- **Treasure Hunt:** Implemented switchable DFS/BFS search using deque, with boundary checks and visited-node tracking to ensure correctness and efficiency.
- **Zombie Defense:** Applied priority queues with custom comparators to manage enemy priority and optimize shooting strategy.
- **Zookeeper:** Developed an optimal-tour solver for a zoo layout using branch-and-bound search with MST-based lower bounds and efficient path-cost evaluation to handle a TSP-style routing problem.

SillyQL: Lightweight Relational Database System

2025.03

A simplified SQL database engine supporting core operations such as CREATE, INSERT, DELETE, PRINT, and JOIN, optimized with hash and BST indexing.

- Implemented field-based conditional queries (WHERE) and two-table joins (JOIN) with automatic index usage or temporary hash-map fallback for fast row matching.
- Designed and integrated hash and BST index structures to accelerate PRINT and DELETE under large datasets, improving query efficiency and scalability.

Software & Web Development

VeroEat Food Safety Assistant App

2026.01 – Present

A family-oriented full-stack health assistant application designed to help users avoid allergens and track food recall updates.

- Implemented barcode scanning and ingredient comparison logic to provide real-time allergen alerts based on multi-user profiles, intelligently recommending safe alternatives.
- Engineered a “Virtual Pantry” module that continuously tracks purchase history and integrates with food safety databases to deliver real-time product recall notifications.

Insta485 Full-Stack Social Media Project Series (Python, HTML, CSS, JS)

2025.09

A three-phase, full-stack development project modeling the architecture of real social platforms, emphasizing web systems fundamentals, clean design, and scalable API development.

- **Static Page:** Built static user pages using Jinja2 templates; applied template inheritance to reduce redundant rendering logic.
- **Server-Side Dynamic page:** Developed a Flask backend with SQLite3 supporting user authenti-

cation, post management, comments/like functionality, and secure cookie-based session control.

- **Client-Side Dynamic page:** Implemented React.js frontend using `useState` and `useEffect`; Built a JSON REST API layer to support asynchronous fetch calls for likes, comments, and feed updates, enabling real-time interactions without full page reloads.

Game Development

AnotherMe – Dual-Character Puzzle Game (Unity)

2026.02

An independently developed 2D side-scrolling puzzle game focusing on cognitive coordination.

- Designed the core “mirror operation” mechanic, requiring players to utilize left-right brain coordination to simultaneously control two avatars.
- Programmed custom physics and collision interactions to support complex, dual-state level challenges.

The Legend of Zelda Classic Clone (Unity)

2026.01

A reverse-engineered recreation of the classic action-adventure game with modernized expansions.

- Reconstructed the underlying game logic, including complex level building, dynamic enemy generation, and interactive scene elements.
- Expanded upon the original framework by designing and implementing a “Custom Level” feature and introducing new interactive mechanics to enhance replayability.

TapTap Spotlight GameJam *Cat Not Found* (Godot Engine)

2025.10 – 2025.11

A 21-day online game development competition held in China.

- Designed storyline and narrative structure; wrote script and dialogue; developed dual-ending logic.
- Created pixel-art assets, 2D scenes, and UI components; ensured coherent visual style.
- Contributed to programming, optimized feature feasibility, and delivered a complete demo.
- Managed coordination and maintained development logs on social media for promotion.

Skills

Programming / Tools: C, C++, Python, R, SQL, HTML, CSS, JavaScript, MATLAB; PyTorch, React, Flask, Godot, Unity, LaTeX

Languages: English (TOEFL 111); Japanese (JLPT N3)

Certificates: Coursera: Intro to HTML5; Intro to CSS3; Python Data Structures; **Kaggle:** Intermediate Python; Pandas; Intermediate Machine Learning; Data Visualization